

VISWESWAR SIRISH PARUPUDI

+1 475-348-4238 | ✉ sirish.p@nyu.edu | 🔗 linkedin.com/in/sirish-visweswar | 🌐 itsSirish | 📁 ittssirish.github.io

EDUCATION

New York University, Tandon School of Engineering

Computer Science, M.S. | GPA: 3.85

Brooklyn, NY

May 2026

Birla Institute of Technology and Science Pilani

Electronics and Communication, B.E. | Minor in Computing and Intelligence | GPA: 8.68/10

Hyd, India

May 2024

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, C#, MATLAB, HTML/CSS

Frameworks: React, Django, Express/Node.js, REST APIs, PyTorch, TensorFlow, Keras

Tools/Platforms: AWS (EC2, Elastic Beanstalk, RDS, S3), Docker, Kubernetes, Git, PostgreSQL, Kafka, Spark, NumPy, Pandas, Matplotlib

EXPERIENCE

Researcher | EEG-fMRI Cross Modal Translation, Foundation Models

June. 2025 – Present

Neural Translation and Foundation Models

Neuroinformatics Lab, NYU, NY

- Building multimodal neural foundation models (CBraMod, BrainLM) over 1.1M+ EEG windows (TUH) and large-scale fMRI datasets (UK Biobank, HCP, NATVIEW) to **learn shared latent spaces** across electrophysiology and hemodynamics.
- Utilized cross-modal translators with asymmetric positional encodings, criss-cross attention, and staged Huber-loss training; achieved ROI-level correlations up to **0.83** on NATVIEW held-out subjects for visual decoding tasks.

Graduate Assistant | Courses: Computer Vision, Neuroinformatics

Aug. 2025 – Present

Tandon School of Engineering, New York University

New York, NY

- Facilitated graduate level live coding sessions, and office hours for **80+ students**; providing mentorship and guidance for projects.
- Authored and graded assignments/exams and built **4** Kaggle-style competitions (dataset curation, baselines, and metrics); evaluated submissions and delivered structured feedback across CV topics (recognition, image processing, deep learning).

Research Assistant | LangChain, LLMs, PyTorch

June. 2023 – Dec. 2023

Text Augmentation for NER and Language Modeling - Undergraduate Thesis

Nanyang Technological University, Singapore

- Improved Named Entity Recognition (NER) on the WNUT17 tweets dataset by mitigating noisy-text challenges using entity placeholder replacements and PEGASUS-based paraphrasing to clean text, yielding more accurate NER predictions.
- Prioritized entity retention, fluency, and BERTScore; boosted performance to F1=0.558 (+43% vs baseline). Leveraged **LLaMA 2** for few-shot paraphrase generation via prompt engineering; achieved peak F1=0.581 (**+85%** vs baseline) with 6% entity loss.

Software Engineer Intern | Python, SQL, Web3

May. 2022 – July. 2022

Ethereum Gas Price Prediction

DLT Labs (KNNX), Hyderabad, IN

- Devised a prediction model to forecast Ethereum gas prices using time series data forecasting techniques, with a focus on ARIMA methodologies. Successfully extracted and processed gas price data directly from Ethereum Blockchain APIs.
- Delivered accurate daily and minute-by-minute gas price forecasts through rigorous hyperparameter tuning and model diagnostics, including ACF/PACF analysis, to **minimize MAE by 27%**, contributing to enhanced decision-making for participants.

PUBLICATIONS

A Smartphone-Enabled Deep Learning Approach for Myocardial Infarction Detection Using ECG Traces for IoT-Based

Healthcare Applications, in **IEEE Sensors Letters**, vol. 7, no. 11, pp. 1-4

Python, Tensorflow, TFLite, Android Studio | Nov. 2023

- The model uses a majority voting ensemble approach with five branches, each containing a quantized transfer learning architecture (FP16 and INT8 formats) to achieve high performance while maintaining efficiency on mobile devices for medical health imaging.
- The overall accuracy of the developed ensemble DL-based model has been obtained as **99.34%** using the fivefold CV scheme. The proposed smartphone-app based healthcare system is portable and can be used for **real-time detection of MI** and other heart abnormalities.

SELECTED PROJECTS

🔗 CineMatch – Group Movie Matching & Social Discovery Platform | React, Django, PostgreSQL, AWS

Sept. 2025 – Dec. 2025

- Deployed a **Django + React** movie discovery/tracking platform with personalized recommendations; built community genre rooms and private groups with synchronized swipes, real-time matches, and live chat via **WebSockets + Redis** reducing decision paralysis.
- Managed ZenHub backlog/velocity and shipped MVP/MLP features (indexed **50K+ titles**, <**300ms** chat delivery, dashboards, catalog import/tracking, full-text search); deployed on **AWS EB/EC2** with **RDS** and enforced quality via Travis CI (**90%+** coverage).

🔗 Scalable Chest X-Ray Triage Pipeline with Spark & Med-GEMMA | Kafka, Spark, AWS S3, Python, Docker

Oct. 2025 – Dec. 2025

- Engineered a production-style pipeline for automated radiology report generation with Kafka ingestion, Spark orchestration, and S3 data lake; including adaptive auto-tuning for GPU batch size and worker concurrency.
- Benchmarked stable scaling with **0.57–0.58 imgs/s** throughput, **1.82s** mean inference and **2.1s** p95 latency; projected **~33K images/day** capacity and **0.00007\$** per report (>**99%** cost reduction).

🔗 Det-IGEN: Detectable Watermarking in AI Image Generation | Python, PyTorch, Diffusion Models

Sept. 2024 – Dec. 2024

- Implemented invisible tree ring watermarking system into the generation process of Diffusion Models by modifying the initial noise vector in the Fourier domain, incorporating an invisible pattern (key) that remains robust against various image transformations.
- Demonstrated negligible impact on image quality, with minimal changes compared to non-watermarked images, enabling **detection of AI generated content (AUC and TPR@1%FPR of 1.000)** and maintained high performance under adversarial conditions.